Project Architecture and System Requirements

The Technical Documentation Assistant operates as a modular Retrieval-Augmented Generation (RAG) system.
Minimum hardware specifications require a dual-core 64-bit x86 or ARM CPU, 8 GB of RAM, and 5 GB of free di
The primary embedding model utilized is all-MiniLM-L6-v2 which maps text chunks into a 384-dimensional dense
For vector similarity indexing, FAISS IndexFlatIP is configured with L2 normalization to compute exact cosine sim

Security, Guardrails, and Data Storage Policies

All uploaded documents and generated vector embeddings are stored in-memory during active sessions.
The system enforces a strict anti-hallucination policy: if the query cannot be answered from retrieved passages, t
Optical Character Recognition (OCR) using Tesseract is recommended if scanned image PDFs without digital te